PROJECT SYNOPSIS

TRUSTLENS: INTELLIGENT COMMENT SENTIMENT & MISLEADING FEEDBACK ANALYSER

P164-HN7:MCA (HONS.)(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

SUBJECT

PROGRAMMING IN JAVA

CAP477

GROUP 2

SECTION - D2533

SUBMITTED BY

JERIN MATHEW VINU (12504196)

ABHIRAM C DINAKARAN (12511639)

SHUBHAM VIJAY YEWALE (12518309)

ASHU (12524062)

SUBMITTED TO

PROF . ANIL SHARMA

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INTRODUCTION

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In today's digital era, the rapid growth of online communication has significantly increased the amount of textual content generated through websites, social platforms, and e-consultation systems. Public participation in decision-making, policy review, and service evaluation has become more common through digital platforms where users can submit comments, feedback, suggestions, and opinions. At the same time, online information sharing has also led to the widespread circulation of misleading, exaggerated, or false content. As a result, organizations and administrators face two major challenges: understanding large volumes of user feedback and identifying unreliable or misleading textual information.

Manual analysis of such textual data is often difficult, time-consuming, and prone to inconsistency. When hundreds or thousands of comments and text entries are submitted, reviewing them individually becomes inefficient. Important opinions may be overlooked, repetitive comments may increase redundancy, and misleading or false statements may affect the overall reliability of the information being analyzed. This creates a strong need for an intelligent automated system that can process textual content efficiently and generate meaningful insights.

TrustLens: Intelligent Content Analysis System Using Machine Learning is proposed as a smart web-based solution designed to address these challenges. The system combines two major analytical modules within a single platform. The first module focuses on Sentiment Analysis of comments received through the E-Consultation module, where user comments are classified as positive, negative, or neutral to help understand user opinion and satisfaction. The second module focuses on Fake News / Misleading Content Detection, where textual news or informational content is analysed to determine whether it is reliable or misleading. These intelligent modules help automate the analysis of textual input and reduce the burden of manual review.

In addition to classification, the system can support efficient content handling, structured storage, and better interpretation of user-submitted information. By integrating machine learning techniques into a Java-based web application, TrustLens transforms unstructured text into useful and actionable insights. The proposed system is especially useful in environments where both public feedback analysis and content reliability verification are important. Thus, TrustLens contributes to improved digital communication, better stakeholder understanding, and more informed decision-making.

LITERATURE REVIEW

LITERATURE REVIEW

The rapid growth of online communication platforms has generated a vast amount of textual data in the form of reviews, comments, opinions, feedback, and news content. This has led to the emergence of text mining, sentiment analysis, opinion mining, and fake news detection as important areas of research in computer science, artificial intelligence, and data analytics. These techniques are widely used to automatically process, analyze, and classify textual information in order to extract meaningful insights from unstructured data.

Sentiment analysis is a computational technique used to identify and classify the emotional tone or opinion expressed in textual content. It is commonly applied in product review analysis, customer feedback systems, social media monitoring, and public opinion studies. Traditional sentiment analysis systems generally classify text into positive, negative, or neutral categories using methods such as lexicon-based analysis, machine learning algorithms, and hybrid approaches. Several studies have shown that sentiment analysis can significantly reduce the manual effort required to process large volumes of user feedback and improve decision-making by highlighting overall opinion trends.

At the same time, the increasing spread of misleading and false information across digital platforms has made fake news detection another important research area. Fake news detection focuses on identifying whether a given piece of textual information is genuine or misleading by analyzing linguistic patterns, writing style, and semantic features. Machine learning techniques such as Naïve Bayes, Support Vector Machine (SVM), Logistic Regression, Random Forest, and TF-IDF-based text classification have been widely used for this purpose. These methods help in automatically detecting suspicious or unreliable textual content and are especially useful in online environments where misinformation can spread rapidly.

Recent research has also highlighted that many existing sentiment analysis systems focus only on emotional polarity and often fail to consider the credibility, misleading nature, or repetitive behavior of user-generated content. In contexts such as e-consultation platforms, feedback systems, and public communication portals, these limitations can reduce the reliability of analysis. Comments containing exaggerated language, unsupported claims, repeated patterns, spam-like structures, or emotionally manipulative expressions may distort the actual understanding of public opinion. Therefore, modern

intelligent text analysis systems increasingly combine sentiment classification with credibility-oriented content evaluation to improve the quality and trustworthiness of interpretation.

In addition to classification, keyword extraction, text summarization, and word cloud generation are commonly used in modern textual analysis systems to provide concise and visual summaries of large text collections. These techniques help analysts quickly identify dominant terms, recurring concerns, common discussion points, and thematic patterns within user comments or textual datasets. Such features improve the interpretability of machine learning outputs and support faster analysis of large-scale text data.

The proposed system, TrustLens, is inspired by these developments and aims to provide a more comprehensive intelligent content analysis solution. Unlike conventional systems that focus on only one type of textual classification, TrustLens integrates Sentiment Analysis for comments received through the E-Consultation module and Fake News / Misleading Content Detection within a unified platform. It also supports efficient interpretation of textual data by enabling structured analysis of user input. This integrated approach makes the system more suitable for environments where both public feedback understanding and content reliability verification are important. Therefore, TrustLens contributes to the growing field of intelligent text analysis by combining multiple machine learning-based approaches in a single web-based application.

FINDINGS AND CONCLUSION

FINDINGS

The development and implementation of the TrustLens system provided several important insights into handling textual data, analyzing user-generated comments, and identifying misleading or false content. The system successfully integrates a user-friendly frontend with a robust backend using Spring Boot and MySQL, enabling efficient storage, retrieval, and processing of user inputs and analysis results. One of the key findings of the project is that textual content received through different modules can be effectively analyzed using machine learning and structured processing techniques to derive meaningful classifications and insights.

The project demonstrates that sentiment analysis can be successfully applied to comments received through the E-Consultation module, allowing the system to classify user feedback into positive, negative, or neutral categories. This helps in understanding user opinions, satisfaction levels, and general response trends in a faster and more organized manner. Another important finding is that textual content such as news or informational input can also be analyzed for fake or misleading characteristics, enabling the system to support reliability verification and reduce the spread of untrustworthy content.

The separation of major modules such as authentication, E-Consultation, fake news detection, sentiment analysis, result management, and admin control improves the overall organization, maintainability, and scalability of the system. The implementation of role-based access for User and Admin enhances both security and usability. Users are able to register, log in, submit textual input, and view analysis results seamlessly, while the admin can monitor users, submitted data, and generated outputs.

The database design, including structured tables for users, comments, consultation data, news inputs, and analysis results, supports proper data management and efficient query execution. During development, practical issues such as API integration errors, data flow mismatches, navigation problems, and communication between backend and ML modules were identified as important factors affecting system performance. These findings highlight the importance of proper testing, debugging, and module integration in the successful development of intelligent web-based applications.

CONCLUSION

The TrustLens project successfully achieves its objective of developing an intelligent content analysis system capable of processing and interpreting textual data in a meaningful way. The system effectively combines Sentiment Analysis of comments received through the E-Consultation module with Fake News / Misleading Content Detection, making it a more comprehensive and practical solution for modern digital platforms. By integrating these two analytical capabilities within a single web-based application, the project demonstrates how machine learning can be used to support both user feedback understanding and content reliability verification.

The project also demonstrates the practical application of modern web development technologies such as HTML, CSS, JavaScript, Spring Boot, and MySQL, along with machine learning integration, in building a complete full-stack intelligent system. It highlights the importance of modular design, database management, backend integration, and text classification techniques in developing scalable and efficient software solutions.

Overall, TrustLens serves as an effective and scalable platform for analyzing user comments and verifying textual content. The system reduces manual effort, improves the speed of interpretation, and supports better decision-making through automated analysis. With further enhancements such as advanced NLP models, real-time analytics, multilingual support, improved UI/UX, and more accurate classification techniques, the system can be extended into a more powerful and intelligent platform for digital content analysis and monitoring.

SCOPE OF THE PROJECT

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The scope of the TrustLens: Intelligent Content Analysis System project is centered on building a web-based platform that helps users and administrators analyze textual content, understand user opinions, and identify misleading or false information. In today's digital environment, users frequently depend on comments, feedback, and online content before making decisions. However, such information can often be misleading, biased, spam-like, or factually incorrect. This project addresses these challenges by providing an automated analysis system that combines sentiment evaluation and content reliability verification.

The project allows users to register, log in, submit textual content such as consultation comments or news input, and receive analysis results. The system processes the submitted text and provides outputs such as sentiment classification (positive, negative, or neutral) and fake or misleading content detection results. This helps users and administrators understand whether the content reflects genuine opinion or contains unreliable or deceptive information. The project also includes an admin module, which expands the system's scope beyond a normal user application. The administrator can log in separately and view registered users, submitted inputs, and generated analysis results, enabling monitoring, record maintenance, and overall system control.

From a technical perspective, the scope includes the integration of:
Frontend using **HTML**, **CSS**, and **JavaScript**
Backend using **JAVA** (**Spring Boot**)
Database using **MySQL**
Machine learning integration using **Python** for text classification and analysis

The project demonstrates the practical implementation of a full-stack web application with authentication, data storage, API communication, and intelligent result generation. The current scope is limited to text-based sentiment analysis and fake news detection and does not yet include advanced deep learning models, image-based content verification, or multilingual analysis. However, the system provides a strong foundation for future expansion into more advanced and scalable intelligent content analysis platforms.

Thus, the scope of TrustLens lies in improving content understanding, assisting users in identifying misleading information, and providing administrators with effective tools for monitoring user feedback and textual data.

METHODOLOGY AND TOOLS

METHODOLOGY

The methodology adopted in the TrustLens system focuses on processing and analyzing textual data using a combination of web technologies and machine learning techniques. The system is designed to handle user-generated content from different modules, such as E-Consultation comments and news or informational text, and classify them based on sentiment and reliability. The process begins with data input, where users submit comments or textual content through the web interface. This data is then received by the Java-based backend system and stored in the database for further processing. Once the input is captured, it is passed to the analysis component, where preprocessing steps are applied. These steps include text cleaning, removal of special characters, conversion to lowercase, and elimination of stop words to improve the quality of analysis.

After preprocessing, the system performs feature extraction, typically using techniques such as TF-IDF (Term Frequency–Inverse Document Frequency) to convert textual data into numerical form suitable for machine learning models. The processed data is then passed to classification models designed for two main purposes: sentiment analysis and fake news or misleading content detection. For the sentiment analysis module, the system classifies the text into categories such as positive, negative, or neutral, helping to understand user opinions and feedback trends. For the fake news detection module, the system evaluates the reliability of the text and predicts whether it is genuine or misleading based on learned patterns.

The results generated by the machine learning models are sent back to the backend, where they are stored and displayed to the user or administrator through the frontend interface. The system also ensures proper role-based access control, allowing users to interact with the application while administrators monitor and manage the overall system.

TOOLS AND TECHNOLOGIES USED

The TrustLens system is developed using a combination of frontend, backend, database, and machine learning tools, ensuring a complete full-stack implementation.

Frontend Technologies

- **HTML** – Used for structuring web pages and forms.
- **CSS** – Used for designing and styling the user interface.
- **JavaScript** – Used for client-side scripting and interactive features.

Backend Technologies

- **Java (Spring Boot)** – Used for developing the backend logic, handling requests, managing APIs, and integrating different modules of the system.

Database

- **MySQL** – Used for storing user data, comments, news inputs, and analysis results in a structured format.

Machine Learning Tools

- **Python** – Used for implementing machine learning models.
- **Scikit-learn** – Used for building classification models such as sentiment analysis and fake news detection.
- **Pandas and NumPy** – Used for data processing and manipulation.

Integration Tools

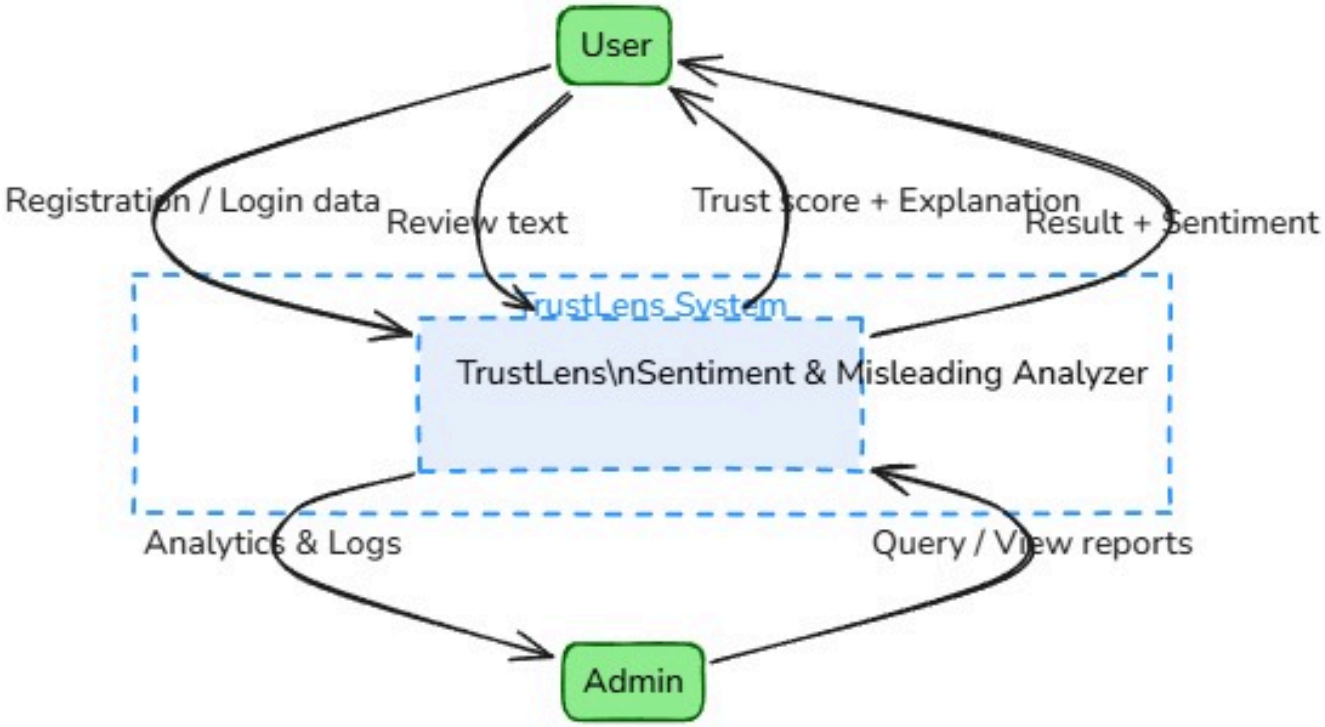
- **REST API / Flask / Script Integration** – Used to connect the Java backend with the machine learning module and exchange data between them.

Development Tools

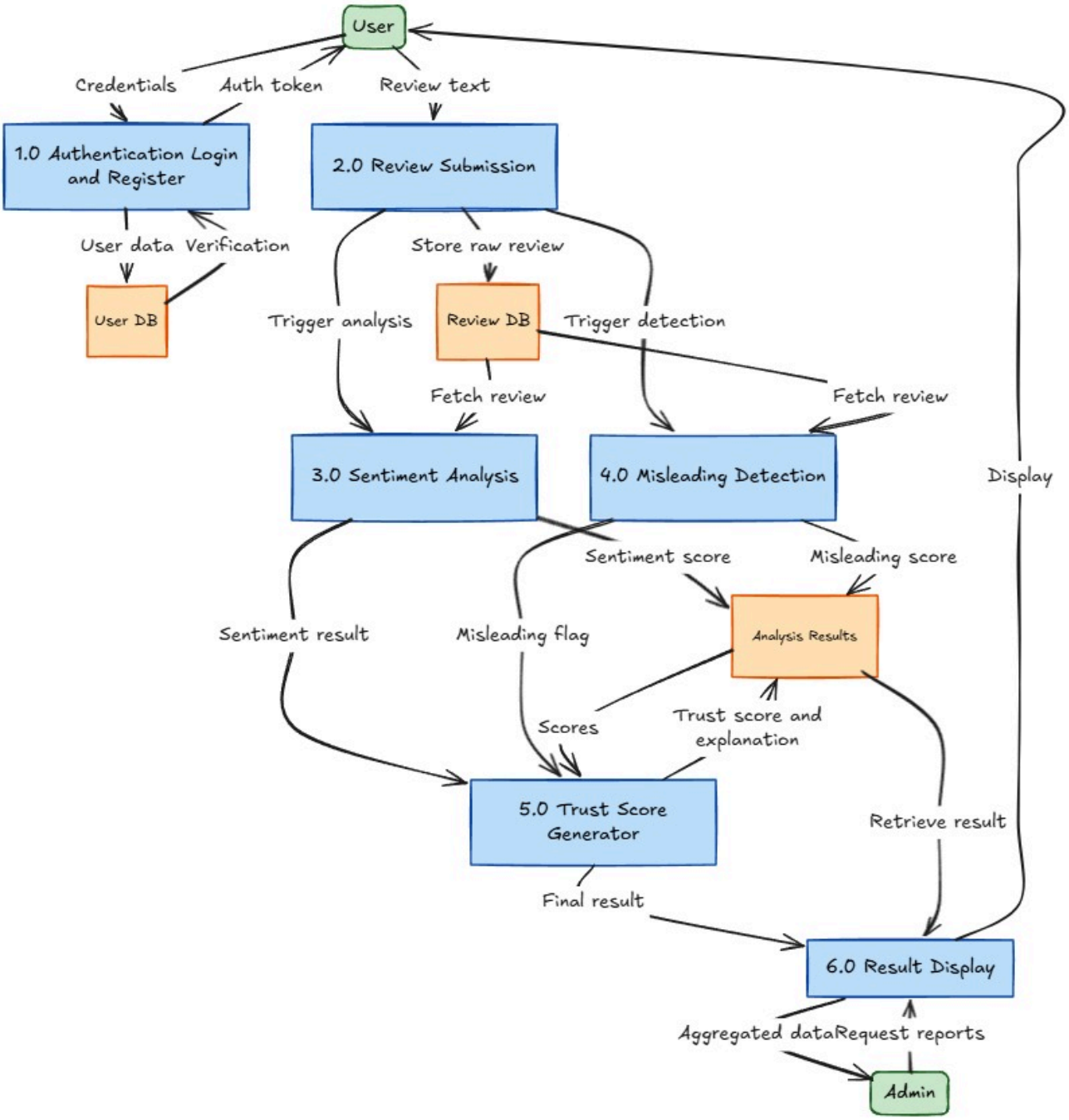
- **IDE (VS Code)** – Used for coding and development.
- **Postman** – Used for API testing and debugging.
- **Git** – Used for version control and project management.

FLOWCHART

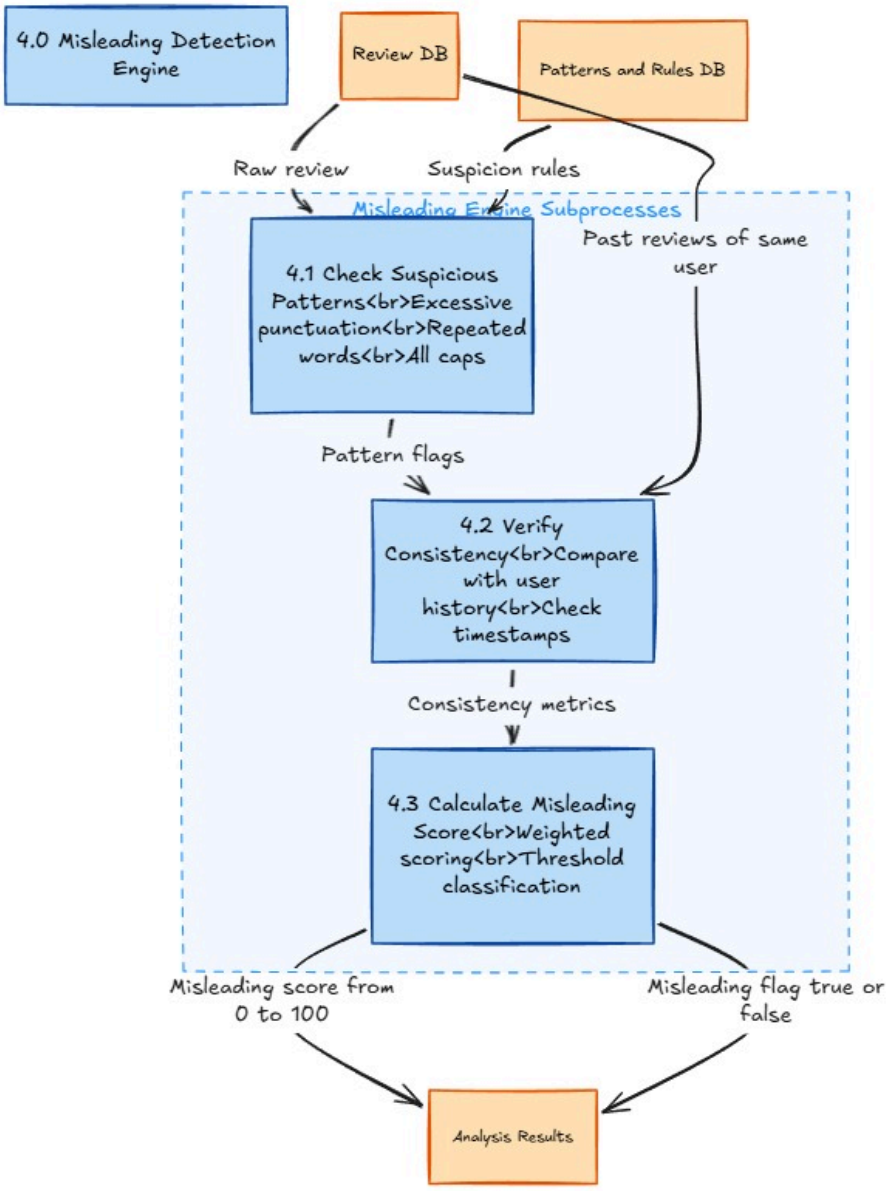
DFD Level 0 (Context Diagram)



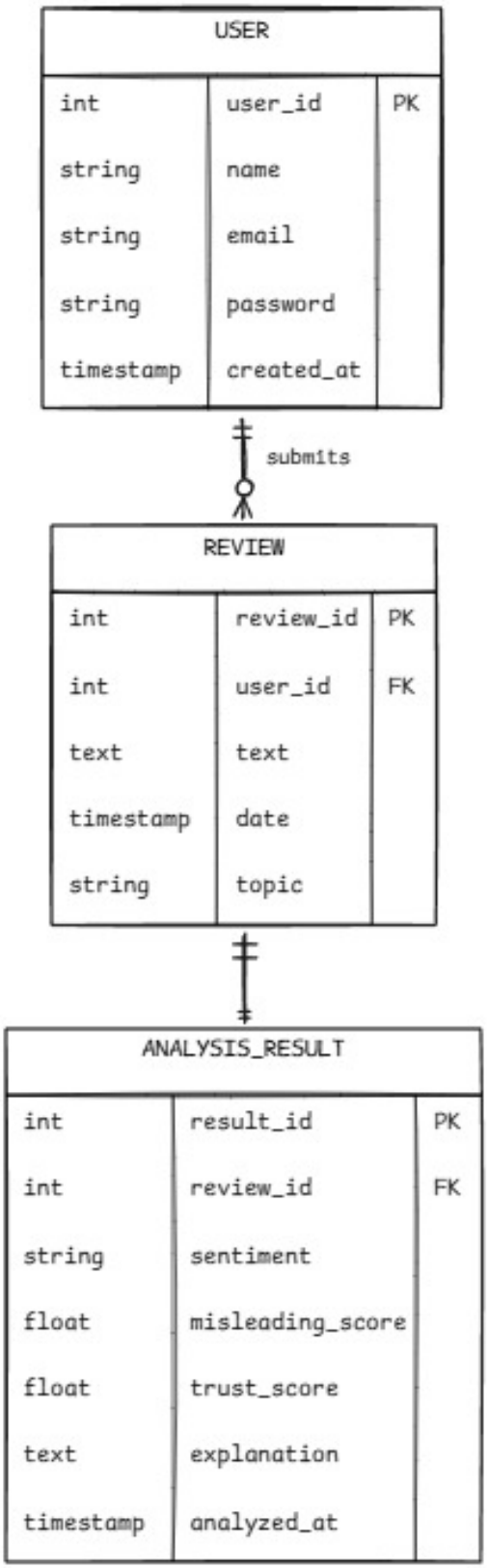
DFD Level 1 (Major Processes & Data Stores)



DFD Level 1 (Major Processes & Data Stores)



ER Diagram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string name
string email
string password
timestamp created_at
submits
REVIEW
int review_id PK
int user_id FK
text text
timestamp date
string topic
ANALYSIS_RESULT
int result_id PK
int review_id FK
string sentiment
float misleading_score
float trust_score
text explanation
timestamp analyzed_at

FEASIBILITY STUDY

FEASIBILITY STUDY

A feasibility study is an important step in system development that helps determine whether a proposed project is practical, achievable, and beneficial. It evaluates the project from different perspectives such as technical feasibility, economic feasibility, operational feasibility, and schedule feasibility. The feasibility study of the TrustLens system shows that the project is realistic and suitable for implementation using available technologies and resources.

1. Technical Feasibility

The TrustLens system is technically feasible because it is developed using widely available and reliable technologies such as HTML, CSS, JavaScript, Spring Boot (Java), MySQL, and Python-based machine learning tools. These technologies are suitable for building a web-based intelligent content analysis platform. The required development environment, software libraries, and frameworks are accessible and can be implemented without requiring highly specialized hardware or expensive infrastructure.

The machine learning modules used for sentiment analysis and fake news detection are also technically feasible, as they can be built using standard tools such as Scikit-learn, Pandas, and NumPy. Since the system mainly processes text-based input, it does not demand high-end computational resources. Therefore, the technical implementation of the project is practical and manageable.

2. Economic Feasibility

The project is economically feasible because it can be developed using mostly open-source software and freely available tools. Technologies such as Java, Spring Boot, MySQL, Python, and machine learning libraries can be used without additional licensing costs. The hardware requirements for development and testing are also minimal, as the system can run on a standard computer or laptop.

Since the project does not require costly infrastructure, special devices, or premium software, the total development cost remains low. This makes the system cost-effective and suitable for academic as well as practical implementation.

3. Operational Feasibility

The TrustLens system is operationally feasible because it is designed to be simple, user-friendly, and practical for both users and administrators. Users can easily register, log in, submit comments or textual content, and receive analysis results. Administrators can monitor submitted data, view analysis outputs, and manage system records efficiently.

The system supports real-world usage by helping users understand feedback sentiment and detect misleading content automatically. Since the interface is easy to use and the functions are clearly organized, the system can be operated without requiring advanced technical knowledge. This makes it suitable for deployment in environments where textual data analysis is needed.

4. Schedule Feasibility

The project is also feasible in terms of time. The system is developed using a modular approach, where different components such as user authentication, E-Consultation, fake news detection, sentiment analysis, and admin management can be built and tested separately. This allows the project to be completed within the available development period.

Since the tools and technologies used are well-documented and widely supported, development and debugging can be carried out efficiently. Therefore, the project is achievable within the planned academic schedule.

Conclusion of Feasibility Study

Based on the above analysis, the TrustLens system is technically, economically, operationally, and schedule-wise feasible. The project can be successfully implemented using available tools, resources, and time. Hence, the proposed system is a practical and viable solution for intelligent textual content analysis.

EXPECTED OUTCOME

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The expected outcome of the TrustLens project is the successful development of a web-based intelligent content analysis system capable of processing and analyzing textual data efficiently. The system is expected to provide users and administrators with meaningful insights by combining sentiment analysis and fake news or misleading content detection within a single platform.

One of the major expected outcomes is the ability of the system to analyze comments received through the E-Consultation module and classify them into positive, negative, or neutral categories. This will help in understanding user opinions, satisfaction levels, and overall response trends more effectively. By automating this process, the system is expected to reduce the time and effort required for manual review of large volumes of comments.

Another important expected outcome is the identification of fake, misleading, or unreliable textual content. The system is expected to analyze user-submitted news or informational text and determine whether the content appears genuine or suspicious. This feature will help improve content reliability and support better decision-making by reducing the impact of misinformation.

The project is also expected to provide a user-friendly interface for both users and administrators. Users will be able to register, log in, submit textual input, and view results, while administrators will be able to monitor submitted data, manage records, and review generated outputs. This is expected to improve system usability, control, and data organization.

Overall, the expected outcome of TrustLens is the creation of an efficient, reliable, and scalable intelligent system that supports text classification, user feedback understanding, and misleading content detection in a practical and organized manner.

SYSTEM DESIGN

1. INTRODUCTION

System design is an important phase in software development that defines the structure, components, and workflow of the proposed system. It provides a clear blueprint for how the application is organized and how different modules interact with each other. The system design of TrustLens: Intelligent Content Analysis System Using Machine Learning is focused on building a structured, scalable, and user-friendly web application capable of analyzing textual data efficiently.

The proposed system is designed to support two major analytical functions: Sentiment Analysis of comments received through the E-Consultation module and Fake News / Misleading Content Detection. The system enables users to submit textual content and receive meaningful analysis results, while administrators can monitor, manage, and review the generated outputs.

The design follows a modular architecture that integrates frontend technologies, backend logic, database management, and machine learning components. This structure ensures that the system remains organized, maintainable, and efficient in handling user interactions and analytical operations. The system design also emphasizes proper input handling, meaningful output generation, and structured data storage.

2. MODULARIZATION

The TrustLens system is divided into different modules to ensure clarity, maintainability, and smooth functionality. Each module is responsible for performing a specific task within the system.

2.1 User Authentication Module

This module handles user registration, login, and access control. It ensures that only authorized users can access the system and submit textual content. It also supports role-based access for users and administrators.

2.2 E-Consultation Module

This module allows users to submit comments, opinions, and feedback through the consultation interface. The submitted comments are stored in the database and later analyzed by the sentiment analysis component.

2.3 Fake News Detection Module

This module is responsible for analyzing textual content such as news statements or informational input. It predicts whether the submitted content is genuine or misleading based on predefined logic and machine learning techniques.

2.4 Sentiment Analysis Module

This module processes comments received through the E-Consultation module and classifies them into **positive, negative, or neutral** categories. It helps in understanding user opinions and feedback trends.

2.5 Analysis Result Module

This module stores and displays the outputs generated by the system, including sentiment labels, trust scores, misleading scores, and explanations. It acts as the bridge between analysis processing and user/admin display.

2.6 Admin Module

This module provides administrative functionalities such as viewing registered users, submitted textual inputs, and generated analysis results. It helps in system monitoring and data management.

3. INPUT DESIGN

Input design refers to the method through which data is entered into the system. A good input design ensures that data is entered accurately, efficiently, and in a user-friendly manner. In the TrustLens system, input design plays an important role because the system depends on textual data provided by users for analysis.

The system accepts input mainly in the form of **user registration details** and **textual content**. During registration, users are required to enter basic details such as name, email, and password. For analytical functions, users can submit textual reviews, comments, consultation feedback, or news-related content through the available forms.

The input forms are designed with simplicity and usability in mind. Validation mechanisms are included to ensure that required fields are not left empty and that incorrect or incomplete data is

minimized. This improves the reliability of the analysis process and prevents unnecessary system errors.

Main Inputs of the System

- User name
- Email address
- Password
- Review / Comment text
- News / Informational text

Thus, the input design of TrustLens ensures smooth data collection and supports effective processing by the backend and machine learning modules.

4. OUTPUT DESIGN

Output design refers to the way in which processed information is presented to the users and administrators. A well-designed output system should be clear, meaningful, and easy to understand. In the TrustLens system, output design is especially important because the main objective of the application is to provide users with analytical insights based on textual data.

The system generates outputs in the form of **analysis results**. For comments received through the E-Consultation module, the system provides **sentiment classification** such as **positive, negative, or neutral**. For textual content related to fake news detection, the system provides results indicating whether the content is **genuine or misleading**.

In addition to classification, the system also displays supporting outputs such as:

- **Trust Score**
- **Misleading Score**
- **Short Explanation of Result**

These outputs help users and administrators better understand the nature of the submitted content. The results are displayed through the web interface in a simple and readable format, ensuring effective communication of the system's findings.

Main Outputs of the System

- Sentiment label
- Fake / Real classification
- Trust score
- Misleading score
- Explanation of result

Thus, the output design of TrustLens is aimed at making analytical results understandable, informative, and useful for decision-making.

5. DATABASE DESIGN

Database design is an important part of system design because it defines how data is stored, organized, and managed within the application. The TrustLens system uses a **relational database** implemented using **MySQL**. The database is designed to store user details, textual inputs, and generated analysis results in a structured and efficient manner.

The system follows a simple but effective database structure in which user information, text submissions, and analysis outputs are stored in separate but related tables. This design reduces redundancy, improves data consistency, and supports efficient retrieval of information whenever needed.

The database also supports **role-based access**, allowing the system to differentiate between normal users and administrators. Proper relationships are maintained between tables using **primary keys** and **foreign keys**, ensuring data integrity and smooth interaction between modules.

The main database entities in the TrustLens system include:

- Users
- Reviews / Comments
- Analysis Results
- Admin Details

Thus, the database design provides a reliable foundation for storing and managing the data required for sentiment analysis and fake news detection.

6. DATABASE TABLES

The TrustLens system contains the following database tables:

6.1 Users Table

The **users** table stores the details of all registered users in the system.

Field Name	Data Type	Description
user_id	INT (PK, AI)	Unique ID of the user
full_name	VARCHAR(100)	Name of the user
email	VARCHAR(100)	Unique email address
password	VARCHAR(255)	Password of the user
role	VARCHAR(20)	Role of the user (User/Admin)
created_at	TIMESTAMP	Account creation date and time

6.2 Reviews Table

The **reviews** table stores textual input submitted by users. This includes comments received through the E-Consultation module and other user-provided textual content.

Field Name	Data Type	Description
review_id	INT (PK, AI)	Unique review ID
user_id	INT (FK)	References the user who submitted the text
review_text	TEXT	User submitted textual content
submitted_at	TIMESTAMP	Date and time of submission

6.3 Analysis Results Table

The **analysis_results** table stores the results generated after analyzing the submitted textual content.

Field Name	Data Type	Description
result_id	INT (PK, AI)	Unique result ID
review_id	INT (FK)	References the related review
sentiment	VARCHAR(50)	Sentiment result (Positive/Negative/Neutral)
misleading_score	VARCHAR(10)	Percentage of misleading nature
trust_score	VARCHAR(10)	Percentage of trust level
explanation	TEXT	Explanation of the analysis result
analyzed_at	TIMESTAMP	Date and time of analysis

6.4 Admin Table

The **admins** table stores administrator details for system management and control.

Field Name	Data Type	Description
admin_id	INT (PK, AI)	Unique admin ID
admin_name	VARCHAR(100)	Name of the admin
email	VARCHAR(100)	Admin email address
password	VARCHAR(255)	Admin password

CONCLUSION

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The system design of TrustLens: Intelligent Content Analysis System plays a vital role in defining the overall structure, functionality, and workflow of the proposed application. A well-planned system design is essential for ensuring that all modules of the system work together in a smooth, organized, and efficient manner. In this project, the design has been developed in such a way that it supports both Sentiment Analysis of comments received through the E-Consultation module and Fake News / Misleading Content Detection within a single integrated platform.

The modular structure of the system improves clarity and maintainability by dividing the application into separate functional units such as user authentication, E-Consultation, fake news detection, sentiment analysis, result generation, and admin management. This modularization makes the system easier to develop, test, manage, and expand in the future. Each module performs a specific function while maintaining smooth interaction with other parts of the system.

The input design ensures that user data and textual content are collected in a simple, user-friendly, and reliable manner. Similarly, the output design is structured to present meaningful analysis results such as sentiment labels, trust scores, misleading scores, and explanations in a clear and understandable format. This helps users and administrators interpret the system's findings easily and make better decisions based on the generated outputs.

The database design further strengthens the system by providing a structured method for storing user details, submitted textual inputs, and analysis results. The use of properly related database tables ensures data consistency, integrity, and efficient retrieval of information. The centralized storage of reviews and their corresponding analysis outputs supports both analytical modules without creating unnecessary complexity.

Overall, the system design of TrustLens successfully provides a strong foundation for the implementation of an intelligent web-based content analysis platform. It combines frontend design, backend processing, database management, and machine learning integration into a unified and practical framework. Therefore, the proposed design is not only suitable for the current project requirements but also flexible enough to support future enhancements and advanced analytical features.